

BACHELOR'S THESIS DEFENSE

AI-ASSISTED MASTER PROMPT GENERATION CHALLENGES AND PROPOSED SOLUTIONS

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17 August 2026

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Motivation & Problem

What Is a Master Prompt?



- Requirements evolve across prompts through additions, modifications, replacements, and removals.
- A master prompt captures the user's currently valid intent — not the entire conversation history.
- It consolidates the evolving requirements into one self-contained specification that can be reused across sessions, agents, or developers.
- It keeps only final active requirements and remove all unsupported details.

DEFINITION

A **master prompt** is a single, self-contained specification of the user's currently valid intent.

A Master Prompt Is Not Exactly a Conversation Summary

“Build a **single-page** portal homepage **as a static site using only plain HTML, CSS, and vanilla JavaScript (three files: index.html, style.css, app.js) — no frameworks or build step.** Layout & navigation: use a horizontal top navigation bar (no sidebar) **with a brand/logo on the left and the nav buttons on the right.** Include exactly four navigation buttons: Dashboard, Policies, History, Contact. **Clicking a button switches between four corresponding in-page views without reloading; the active button is highlighted.**”

Error 1 — Unsupported details (extrinsic hallucination) (bold): details introduced during consolidation that are not supported by the user’s prompt history.

Error 2 — Inactive constraint (underlined): the sidebar was removed earlier, but stating “no sidebar” not an active requirement in the ground-truth Master Prompt..



Claude · simple project · basic consolidation — AlignScore 0.086, 92% untraceable

From Prompt History to a Master Prompt



- “Create a portal homepage.”
- “Add 3 navigation buttons: Home, Orders, Contact.”
- “Add sidebar navigation.”
- “Undo sidebar.” (undo)
- “Use top navigation instead.” (replace)
- “Change navigation buttons: Dashboard, Policies, History, Contact.” (replace)



consolidate



Expected Master Prompt

“Create a portal homepage. Add top navigation. The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”

A master prompt is not a summary of everything the user said — it is a specification of what is still valid at the end of the history.

Properties of a Reliable Master Prompt

- **Faithful** — every requirement is supported by the user's prompt history; unsupported additions or extrinsic hallucinations are excluded.
- **Current** — later modifications, removals, and replacements are reflected in the final specification.
- **Conflict-aware** — if a later instruction conflicts with a settled requirement without clearly requesting a revision, surface the conflict instead of silently changing the requirement.

Research Questions & Contributions

Research Questions

- **RQ1** — To what extent do AI-generated master prompts contain non-traceable content, including agent-added and outdated requirements, and how does consolidation instruction strictness affect this problem?
- **RQ2** — Does spec-guided consolidation produce a more traceable master prompt than single-pass consolidation of the raw prompt history?
- **RQ3** — When a later instruction semantically conflicts with a settled requirement, how can a spec-guided approach surface the conflict, trace both sides, and avoid resolving it without user input?

- **Requirement-level traceability evaluation** — a requirement-level method using atomic decomposition and entailment-based AlignScore to evaluate whether candidate master-prompt requirements are supported by the ground truth.
- **Spec-guided consolidation** — a maintained specification is updated throughout the prompt history rather than reconstructed only at the end.
- **Conflict-aware project handoff** — later instructions are checked against the settled specification; unclear conflicts are preserved for user resolution rather than silently resolved.

Related Work

Related Work

- **Mondal et al. (2024)** — *Enhancing User Interaction in ChatGPT* — combines sequences of prompts into one prompt; the main goal is interaction efficiency. Does not evaluate requirement-level traceability or later undo/replacement behavior.
- **AlignScore / FactScore / Claimify** — provide ideas for semantic support and atomic evaluation. Do not construct or maintain a master prompt.
- **Spec-driven development** — treats a specification as a maintained artifact. Does not directly evaluate unsupported or outdated requirements in an AI-generated master prompt.

Closest Prior Work: Semantic Commit

- **Semantic Commit** checks new information for semantic conflicts before updating an intent specification and can suggest resolutions for user review.
- **My thesis:** studies the construction of a Master Prompt that remains traceable to the original prompt history.
- For unclear conflicts, the settled requirement is preserved, trace back to requirement that cause conflicts, and the user is asked to decide rather than automatically choosing or merging them.

Methodology

Experimental Design: Two Stages of the Master-Prompt Lifecycle

Master-Prompt Construction *“Can a traceable master prompt be constructed from an evolving prompt history?”*

- **Input:** 2 project histories
- **Process:** 4 consolidation strategies $\times$ 3 AI coding agents
- **Output:** candidate master prompts
- **Evaluation:** requirement-level traceability (AlignScore + untraceable rate)

Semantic Conflict Handling *“Can later semantic conflicts be surfaced without silently changing a settled master prompt?”*

- **Input:** hand-written ground-truth master prompt as settled specification + later prompts
- **Process:** 3 conflict-handling conditions $\times$ the same 3 AI coding agents
- **Output:** updated specification / conflict-handling behavior
- **Evaluation:** qualitative review (RQ3)

Experiment 1 uses ground truth as a scoring reference. Experiment 2 reuses it as the settled starting specification.

Two Controlled Project Histories and Ground Truth

Project	Prompts	Description
Simple	12	Portal with navigation, conditional styling, and contact form
Hard	45	ERP-style application with customers, orders, invoices, roles, and permissions; more revisions and removals

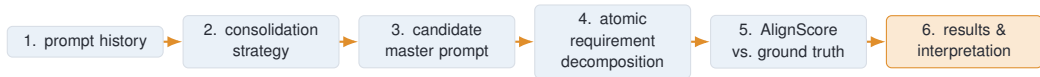
GROUND TRUTH

Manually constructed from the ordered history: retains only requirements still valid at the end; later modifications replace earlier values; removals/undo remove the affected requirement; unsupported details are excluded.

For Experiment 1, the ground-truth master prompt is also manually decomposed into atomic requirements.

Experiment 1: Traceability Evaluation

Experiment 1 Pipeline



2 histories × 3 agents × 4 strategies = 24 candidates.

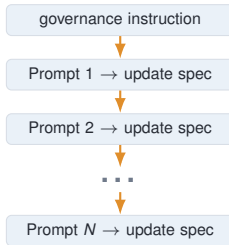
Four Consolidation Strategies

Group A — Single-pass $P_1 \rightarrow \dots \rightarrow P_N \rightarrow$
consolidate once

- **Basic** — consolidate the conversation once at the end.
- **Loose** — additionally account for modifications, removals, and missing/additional features.
- **Strict** — additionally require every requirement to be traceable to the prompt history.

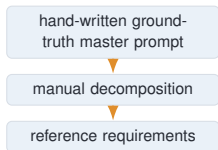
Group B — Per-turn $P_1, P_2, \dots, P_N \rightarrow$
update spec after each

- **Spec-guided** — before the prompt history begins, give the agent a governance instruction to maintain `prompts.md` and `spec.md`; after each new prompt, update the current specification.

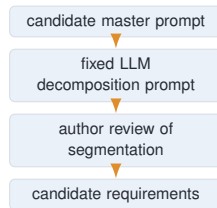


Challenge: Defining the Evaluation Unit

Ground truth



Candidate



Example: “Add 4 navigation buttons: Dashboard, Policies, History, Contact.” is kept as **one** requirement, not split into four — splitting would lose the four-button-set constraint.

Atomic means a meaningful software-requirement unit, not the smallest possible sentence.

Why AlignScore Is Needed for Traceability

Pair 1 — Contradiction

GROUND TRUTH

“A navigation label starting with H must have a blue background.”

CANDIDATE

“A navigation label starting with H must have a red background.”

BERTScore (P): **0.98** AlignScore: **0.0002**

Pair 2 — Faithful paraphrase

GROUND TRUTH

“Implement customer management with create, edit, delete, and browse operations.”

CANDIDATE

“Include customer management: Users can create, edit, delete, and browse customers.”

BERTScore (P): **0.63** AlignScore: **0.87**

`context = ground-truth master prompt,    claim = candidate requirement`

Similar wording can still contradict the requirement, while different wording can express the same requirement.

RQ1 — Effect of Consolidation Instruction Strictness

Project	Agent	basic	loose	strict
simple	Claude	0.086 (92%)	0.824 (10%)	0.988 (0%)
simple	Codex	0.387 (62%)	0.857 (18%)	0.828 (18%)
simple	Gemini	0.083 (97%)	0.737 (21%)	0.917 (10%)
hard	Claude	0.382 (62%)	0.362 (67%)	0.905 (3%)
hard	Codex	0.787 (19%)	0.772 (21%)	0.860 (10%)
hard	Gemini	0.391 (68%)	0.414 (60%)	0.865 (10%)

- Strict > Basic in 6/6 comparisons; Strict > Loose in 5/6.
- Improvement from Basic → Loose → Strict is not always guaranteed (e.g. simple project, Codex).

The explicit rule for traceability matters more than simply adding more instructions.

RQ2 — Spec-Guided vs. Single-Pass Consolidation

Project	Agent	strict	spec-guided
simple	Claude	0.988 (0%)	0.989 (0%)
simple	Codex	0.828 (18%)	0.987 (0%)
simple	Gemini	0.917 (10%)	0.989 (0%)
hard	Claude	0.905 (3%)	0.882 (11%)
hard	Codex	0.860 (10%)	0.895 (8%)
hard	Gemini	0.865 (10%)	0.785 (17%)

Spec-guided improves traceability over the same agent's strict baseline in 4 of 6 comparisons. All three agents improve on the simple project; the hard project is mixed. Thus, spec-guided can improve traceability, but not consistently across all projects and agents.

Example: Non-Traceable Content in Single-Pass Consolidation

CODEX · SIMPLE PROJECT · BASIC

“Create a **responsive static** portal homepage **using HTML, CSS, and JavaScript.**”

“Build a **modern** portal homepage, **not a marketing landing page.**”

“**Do not include a dark theme.**”

“Make sure the page works well on mobile screens: **top navigation should wrap or stack cleanly; content should collapse into single-column layouts; avoid text overflow or layout overlap on narrow screens.**”

“Keep the design polished, clean, and portal-like with **dashboard sections, quick actions, status panels, announcements,** and a usable first screen.”

AlignScore: 0.387 · Untraceable rate: 62% (10 of 16 requirements).

Example: Spec-Guided Consolidation

CODEX · SIMPLE PROJECT · SPEC-GUIDED

“Create a portal homepage. Use top navigation with 4 navigation buttons: Dashboard, Policies, History, Contact; style navigation: rounded buttons, blue accent, modern font. If the label of the navigation button starts with ‘H’, the background color of the button should be blue; if the label starts with ‘O’, the background color should be red; if the label starts with ‘D’, the background color should be ‘yellow’. Add a contact form with name, email, and message fields. The email field must be validated before the form can submit. Make sure the page also works well on mobile screens.”

- Never mentions dark theme or sidebar — both were removed. Removed requirements are excluded from the final specification.

AlignScore: 0.987 · Untraceable rate: 0%.

Challenge: Updating Specification Per Turn

Problem — keeping `spec.md` a self-contained, current specification

- **Edited wording** can remain in `spec.md`: The earlier prompt is “Add 3 navigation buttons: Home, Orders, Contact” and then later “Change navigation buttons: Dashboard, Policies, History, Contact” → should become “Add 4 navigation buttons: . . .”
- **Conversation-like wording** can remain in `spec.md`: “I’d like to build a simple ERP web application for a small business” → should become “Build a simple ERP web application for a small business”

WHY IT HAPPENS

Per-turn updates can preserve the wording of the latest instruction instead of reconstructing the final requirement.

DESIGN RESPONSE

Reconstruct the affected topic from its relevant prompt history instead of patching the previous specification sentence.

Experiment 1 — Interpretation and Trade-Off

What worked

- Explicit traceability rules improve single-pass consolidation.
- Strict outperforms Basic in 6/6 comparisons.

What is mixed

- Spec-guided improves in 4/6 comparisons.
- Strong on the simple project, mixed on the hard project.

Why

Single-pass → one large reconstruction decision

Spec-guided → many smaller interpretation decisions

The 45-prompt hard project contains more revisions and removals than the 12-prompt simple project, so repeated per-turn updates create more opportunities for interpretation errors.

Spec-guided should be understood as an alternative workflow, not a guaranteed improvement.

Experiment 2: Semantic Conflict Handling

Experiment 2 Pipeline



5 semantic conflicts + 2 compatible controls.

What Counts as a Semantic Conflict?

Settled master prompt: “Invoice should have a due date **15 days** after they are created. . .”

Clear Revision

EXAMPLE

“Change the due date to 30 days.”

MEANING

The user replaces the settled 15-day value.

ACTION

Update the requirement.

Semantic Conflict

EXAMPLE

“Add a reminder before the 30-day due date.”

MEANING

Assumes 30 days instead of settled 15-day value.

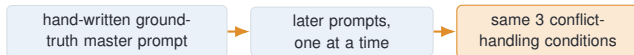
ACTION

Keep the settled requirement and report the conflict for users.

Semantic conflict = an incompatible later assumption without a clear revision request.

Experiment 2: Controlled Prompts Setup

Starting point



Hard project — 4 semantic conflicts

- **Due-date value** — 15-day due date vs. assumed 30-day due date
- **Recycle-bin scope** — vendors-only recycle bin vs. customer recycle-bin tab
- **Navigation layout** — top navigation bar vs. collapsible left sidebar
- **Permission** — staff cannot delete products vs. audit entry for staff deletions

Simple project — 1 semantic conflict

- **Navigation count** — four navigation buttons vs. five buttons wrapping onto two rows

Compatible controls

- Hard: print button
- Simple: footer note

All agents and conditions receive the same settled specification and the same later prompts within each project.

Experiment 2: Conditions and Evaluation

Three Conditions

- **Consolidate-then-ask** — consolidate first, then inspect conflicts afterward.
- **Do-not-resolve** — do not choose or merge conflicting values; keep the conflict separate.
- **Spec-guided** — check each later prompt against the settled specification as it arrives.

Manual Evaluation

- Detect the conflict
- Do not resolve it automatically
- Keep the settled requirement
- Do not invent new details
- Show both sources

Compatible additions should be accepted normally.

Gemini Output: Consolidate-Then-Ask

Settled: “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”

Later: “Make sure all five navigation buttons wrap onto two rows on small screens.”

Later: “Add a footer at the bottom of the page showing a copyright notice.”

Output: Master prompt — “Create a portal homepage. Add top navigation. The navigation must contain **five navigation buttons**, including Dashboard, Policies, History, and Contact. If the label of a navigation button starts with ‘H’, its background color should be blue; if the label starts with ‘O’, its background color should be red; if the label starts with ‘D’, its background color should be yellow. Style the navigation with rounded buttons, a blue accent, and a modern font. Make sure all **five navigation buttons wrap onto two rows** on small screens. Add a contact form with name, email, and message fields. The email field must be validated before the form can submit. Make sure the page also works well on mobile screens. Add a footer at the bottom of the page showing a copyright notice.”

Unresolved conflict

- **Settled / Initial Requirement (Prompt 1):** “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”
- **Later Instruction (Prompt 2):** “Make sure all five navigation buttons wrap onto two rows on small screens.”
- **Conflict Analysis:** a navigation bar cannot contain both four and five buttons at once — the later instruction raises the count from four to five.

Gemini Output: Do-Not-Resolve

Settled: “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”

Later: “Make sure all five navigation buttons wrap onto two rows on small screens.”

Later: “Add a footer at the bottom of the page showing a copyright notice.”

Output: Master prompt — “Create a portal homepage. Add top navigation. If the label of a navigation button starts with ‘H’, its background color should be blue; if the label starts with ‘O’, its background color should be red; if the label starts with ‘D’, its background color should be yellow. Style the navigation with rounded buttons, a blue accent, and a modern font. Make sure navigation buttons wrap onto two rows on small screens. Add a contact form with name, email, and message fields. The email field must be validated before the form can submit. Make sure the page also works well on mobile screens. Add a footer at the bottom of the page showing a copyright notice.”

Unresolved conflict — Number of Navigation Buttons & Button List

- **Prompt 1:** “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”
- **Prompt 2:** “Make sure all five navigation buttons wrap onto two rows on small screens.”

Gemini Output: Spec-Guided Conflict Handling

Settled: “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”

Later: “Make sure all five navigation buttons wrap onto two rows on small screens.”

Later: “Add a footer at the bottom of the page showing a copyright notice.”

Output: Master prompt — Create a portal homepage with top navigation, a contact form, and a footer.

- **Navigation Styling & Behavior:** rounded buttons, blue accent, modern font; H→blue, O→red, D→yellow; buttons wrap onto two rows on small screens.
- **Contact Form:** name, email, and message fields; email validated before the form can submit.
- **Footer:** copyright notice at the bottom of the page.
- **Responsiveness:** page works well on mobile screens.

Unresolved conflict — Number of Navigation Buttons

- **Settled requirement:** “The navigation must contain four buttons: Dashboard, Policies, History, and Contact.”
- **New prompt instruction:** “Make sure all five navigation buttons wrap onto two rows on small screens.”
- **Why:** the initial prompt explicitly specifies four buttons, while a subsequent prompt specifies wrapping “all five navigation buttons” onto two rows without explicitly revising the button count or naming a fifth button.

Experiment 2: Interpretation and Key Challenge

Interpretation

- **Consolidate-then-ask** — the conflicting assumption can enter the Master Prompt before it is checked.
- **Do-not-resolve** — keeps the conflicting value outside the Master Prompt.
- **Spec-guided** — checks the later instruction before updating the settled specification, preserves the settled value, and asks the user.

KEY CHALLENGE — REVISION OR CONFLICT?

A later prompt may introduce a new value without making it clear whether the user intended to revise the settled requirement.

DESIGN RESPONSE

If the revision is unclear, preserve the settled requirement and ask the user.

The main difference is when the conflict is checked: spec-guided checks the new instruction against the settled Master Prompt before changing it.

Conclusion

Conclusion: Answers to the Research Questions

- **RQ1** — Explicit traceability constraints reduce non-traceable content in single-pass consolidation.
- **RQ2** — Spec-guided improves 4/6 comparisons, but replaces one large reconstruction with repeated interpretation decisions.
- **RQ3** — The qualitative conflict study shows that a settled specification can be used to surface incompatible later assumptions without silently resolving them.

TAKEAWAY

A master prompt should be treated as a maintained project specification, not merely as a one-shot summary of prompt history.

Thank you

Time For Questions